Jyothy Institute of Technology
Department of Artificial Intelligence & Machine Learning
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SEVAM

सेवम् | सेवा

A Personalized Ayurvedic Health Companion

Powered by RAG + Local LLM + NLP

PROJECT SYNOPSIS

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1. Abstract

Sevam (Sanskrit: *सेवम्*, meaning *apple*; and *सेवा*, meaning *selfless service*) is a personalized Ayurvedic health companion designed to help everyday users — especially in India — understand their symptoms through the lens of traditional Ayurvedic medicine. Inspired by the timeless principle *"one apple a day keeps the doctor away,"* Sevam aims to be that daily companion: always available, never alarmist, and deeply personal.

Unlike generic internet searches that produce panic and generic results, Sevam knows **who you are** (your Prakriti — dosha constitution), **what you ate** (daily food diary), and **what you feel** (symptom description). By combining these three pillars with a curated Ayurvedic knowledge base and a Retrieval-Augmented Generation (RAG) pipeline, Sevam delivers grounded, personalized, home-remedy-first guidance with zero hallucinations and zero prescription drug suggestions.

The system uses spaCy for NLP symptom extraction, SentenceTransformers for semantic retrieval, ChromaDB as the vector store, FastAPI as the backend, and a locally running LLM via Ollama — ensuring complete privacy and zero recurring API cost.

2. Problem Statement

Consider a common scenario: an elderly person wakes up at midnight with chest pain. He is frightened, unaware of what is happening, and does not know what to do. He searches online and immediately sees results about cardiac arrest and heart attacks. The panic worsens. He may rush to an emergency room unnecessarily — or worse, ignore a real warning sign because he is overwhelmed.

This scenario repeats for millions of Indians daily. The core problems are:

- Generic internet searches do not know the user's diet, health history, or body constitution. They return the worst-case scenario every time.
- Ayurvedic knowledge — thousands of years of holistic, side-effect-minimal remedies — is disappearing from everyday life. Vaidyas treated the same symptoms effectively long before English medicine existed.
- Existing symptom apps are cloud-dependent, impersonal, and often suggest allopathic medicines outside their legal scope.
- No existing tool correlates what you ate today with what you feel tonight — yet food is the most common root cause of everyday symptoms.

Sevam solves this by replacing panic-inducing searches with calm, context-aware, Ayurveda-first guidance that is grounded in retrieved medical knowledge — not guesswork.

3. Objectives

1. Build a RAG-powered symptom assistant grounded exclusively in a curated Ayurvedic knowledge base — eliminating hallucinations.
2. Implement personalized responses based on the user's Prakriti (Vata / Pitta / Kapha dosha profile).
3. Develop a daily food diary with food-symptom correlation — connecting what you ate to what you feel.
4. Integrate a local LLM via Ollama for zero API cost and complete health data privacy.
5. Build a strict safety guardrail layer that detects emergency symptoms and escalates immediately before any LLM call.
6. Deliver a mobile-first, accessible interface usable by non-technical users including the elderly.
7. Strictly avoid suggesting prescription medicines, making diagnoses, or replacing professional medical consultation.

4. Existing Systems & Identified Gaps

Several Ayurvedic and health apps exist in the market today (AyurNod AI, Vaidya AI, AyuRythm, Practo, etc.). A comparative analysis reveals the following critical gaps:

Feature	Existing Apps	Sevam
Food diary + symptom correlation	✗ Absent	✓ Core feature — correlates last 7 days of meals with symptom
Prakriti-based personalization	Partial	✓ Full dosha profile at onboarding, used in every response
RAG-grounded responses (no hallucination)	✗ Absent	✓ Every answer retrieved from KB; LLM only synthesizes
Local / offline LLM (privacy-first)	✗ Cloud only	✓ Ollama on-device — health data never leaves the user
Emergency detection before LLM call	Partial	✓ Rule-based guard fires before any AI processing
Ayurveda-only scope (no prescriptions)	✗ Mixed	✓ Strictly Ayurvedic; legal boundary enforced in prompts
Zero recurring cost	✗ Subscription	✓ Fully open-source stack, no API fees

5. Proposed System

Sevam is built on three foundational pillars that work together to produce a personalized response:

Pillar 1 — Who You Are (Prakriti Profile)

- Dosha assessment via a validated 20-question questionnaire at onboarding
- Stores existing medical conditions, allergies, current medications, sleep and stress patterns
- Every chatbot response is filtered through this profile

Pillar 2 — What You Ate (Food Diary)

- Users log meals in natural language (e.g., "idli chai in morning, biryani and cold drink at night")
- System parses food qualities: hot/cold, heavy/light, spicy/bland, Pitta/Vata/Kapha aggravating
- When a symptom is reported, the last 48 hours of food logs are analyzed for likely dietary triggers

Pillar 3 — What You Feel (Symptom Chat)

- NLP pipeline extracts: intent, symptoms, duration, severity, body part, triggers
- Emergency detector scans for red-flag keywords before any LLM call

- RAG retrieves top-3 relevant Ayurvedic knowledge chunks
- Ollama LLM synthesizes a personalized, grounded response with cited sources

6. System Architecture

Sevam follows a clean, layered architecture separating the mobile frontend, REST API backend, AI/NLP processing layer, RAG retrieval engine, and local LLM inference.

Frontend Layer	Backend Layer	AI / Data Layer
React (Web) PWA / Mobile-first UI Chat Interface Food Diary Health Profile Emergency Overlay	FastAPI (Python) REST Endpoints /chat /food-log /profile /feedback /health-check CORS + Auth Middleware	spaCy NLP Pipeline SentenceTransformers ChromaDB Vector Store Ollama Local LLM MongoDB Atlas (user data) Ayurvedic Knowledge Base

7. System Workflow

The complete end-to-end flow from user input to final response is described below:

Step 1 — User opens Sevam and types symptom (e.g., "I have a headache since morning")

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Step 2 — Safety Guard Checks for emergency keywords (chest pain + radiation, breathing difficulty, unconscious) If EMERGENCY → show emergency card immediately, skip all AI

▼

Step 3 — NLP Pipeline (spaCy) Extracts: Intent | Symptoms | Severity | Duration | Body Part | Triggers

▼

Step 4 — Context Engine Loads: Prakriti Profile + Last 7-day Food Logs + NLP output Builds unified context JSON for retrieval and LLM

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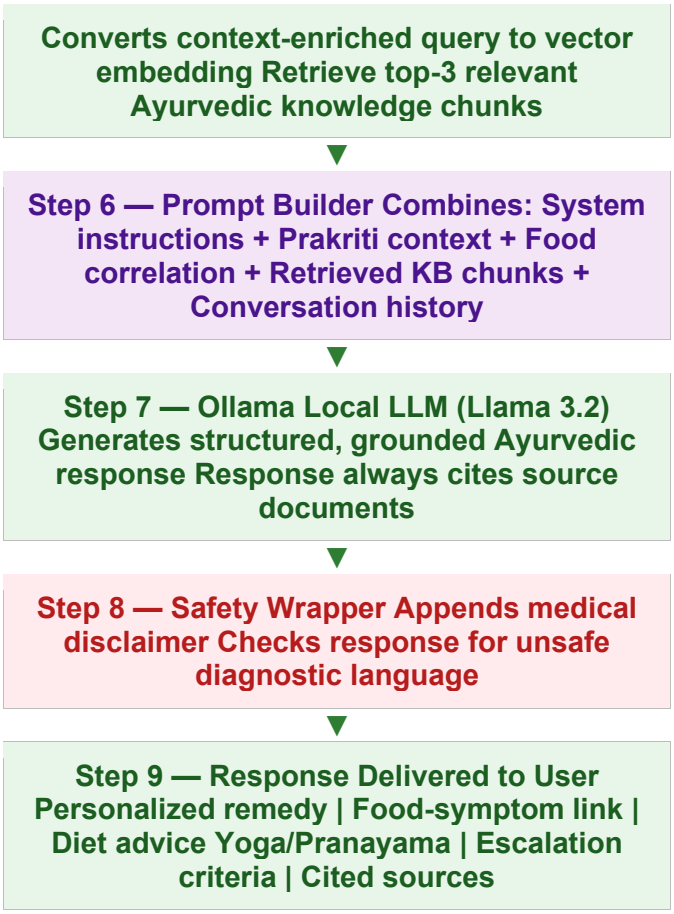
Step 5 — RAG Retrieval (SentenceTransformers + ChromaDB)



8. RAG Pipeline — Detailed Flow

Retrieval-Augmented Generation (RAG) is the core AI technique that prevents hallucinations. Instead of the LLM generating answers from memory, it is forced to answer only from retrieved documents.

Indexing Phase (runs once)	Query Phase (every user message)
1. Curated Ayurvedic documents (Markdown) 2. Text cleaned and chunked (~150 words / chunk) 3. SentenceTransformers generates 384-dim vectors 4. Vectors stored in ChromaDB with metadata (dosha tag, symptom tag, category, source)	**1. User query + context → single enriched query** 2. Embedded with same SentenceTransformers model 3. Cosine similarity search in ChromaDB 4. Top-3 most relevant chunks retrieved 5. Chunks injected into LLM prompt as context

9. Personalization Engine

What makes Sevam fundamentally different is that the same symptom produces a completely different response for different users. This is true to Ayurvedic principles — root cause and remedy depend on the individual, not just the disease.

	User A — Vata Prakriti	User B — Pitta Prakriti
Symptom	Headache since morning	Headache since morning
Food Log	Skipped lunch, 3 hours sleep, stressed	Spicy biryani + cold drink last night
Sevam Response	Warm sesame oil head massage (Abhyanga), Ashwagandha with warm milk, eat warm food immediately, rest in quiet room	Cold coconut oil on scalp, coriander + fennel water, avoid spicy food and screen exposure today

Same symptom. Different root cause. Different remedy. This is what Ayurveda has always known.

10. Technology Stack

Component	Technology	Purpose
Frontend	React + Tailwind CSS	Chat UI, food diary, health profile, emergency overlay
Backend API	FastAPI (Python)	REST endpoints, request validation, middleware
NLP Layer	spaCy	Intent classification, symptom extraction, severity detection
Embeddings	SentenceTransformers (all-MiniLM-L6-v2)	384-dim semantic vectors for RAG retrieval
Vector Database	ChromaDB	Stores and indexes all Ayurvedic document embeddings
Local LLM	Ollama (Llama 3.2 / Qwen2.5)	Local inference — no API cost, full data privacy
Database	MongoDB Atlas	User profiles, food logs, conversations, feedback
Safety Layer	Rule-based Python	Emergency keyword detection before any AI call
Deployment	Vercel (frontend) + Render (backend)	Free-tier cloud deployment

11. Safety Guardrails

Medical AI carries inherent risk. Sevam implements a multi-layer safety architecture:

Layer 1 — Pre-LLM Emergency Guard

- Scans every user message for emergency patterns before any AI processing
- Patterns: chest pain + arm/jaw radiation, breathing difficulty, loss of consciousness, stroke signs
- On detection: returns emergency card instantly with emergency contact numbers (112 / 108 / 911)
- No LLM call is made — speed is critical in emergencies

Layer 2 — Prompt Engineering Safety

- System prompt explicitly instructs LLM: never diagnose, never suggest allopathic drugs
- LLM must answer only from retrieved context chunks (cannot use training memory)
- Temperature set to 0.3 — low creativity, high factual consistency

Layer 3 — Response Wrapper

- Every response gets a mandatory medical disclaimer appended
- Response is scanned for diagnostic language ("you have", "this is definitely") and flagged
- Escalation criteria always included: "If symptoms persist beyond X hours, consult a practitioner"

12. Knowledge Base Strategy

The quality of Sevam's responses depends entirely on the quality of the knowledge base. All documents follow a strict structured format:

- Disease name (English + Ayurvedic terminology)
- Aggravated Dosha
- Symptoms and presentation
- Ayurvedic root cause explanation
- Herbs and preparation method
- Step-by-step home remedy
- Dietary guidance (what to eat / avoid)
- Yoga or Pranayama technique (if applicable)
- Clear escalation criteria — when to seek professional help

Planned knowledge base: 50 documents across Ayurvedic symptom remedies, herb library (turmeric, ginger, ashwagandha, neem, ajwain, tulsi), dosha-based analysis (Vata/Pitta/Kapha imbalances), and yoga/breathing techniques.

13. Future Scope

8. Food Photo Analysis — Camera integration with Ollama Vision model (LLaVA / Llama 3.2 Vision) to automatically identify food items, estimate macros (protein / carbohydrate / fat), and infer Ayurvedic food qualities from a single photo.
9. React Native Mobile App — Native Android and iOS application with push notifications, wearable integration (sleep/stress data), and full offline capability.
10. QLoRA Fine-Tuning — Fine-tune a small LLM (Phi-3 Mini or TinyLlama) on structured Ayurvedic conversation data to produce a model that speaks Ayurveda natively, reducing RAG retrieval dependency.
11. Multi-language Support — Hindi, Kannada, Telugu, and Tamil interfaces making the app accessible to rural and elderly populations.
12. Wearable Integration — Sync with fitness bands for sleep quality, heart rate, and stress levels to automatically enrich the symptom context.
13. Community Knowledge Expansion — Moderated platform allowing Ayurvedic practitioners to contribute and review knowledge base documents.

14. Implementation Roadmap

Phase	Name	Key Deliverables	Timeline
1	**Foundation**	Project structure, Ollama setup, MongoDB Atlas, basic auth	Days 1–10
2	**Core Chat**	Dosha onboarding, RAG pipeline, NLP layer, safety guard, basic chat	Days 11–25
3	**Food Diary**	Natural language food logging, food-symptom correlation engine	Days 26–35
4	**Context Engine**	Unified profile + food + symptom context, personalized responses	Days 36–45
5	**Polish & Deploy**	Full UI, citations, history, PWA, Vercel + Render deployment	Days 46–60

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